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Top Skills

Java
credit risk transfer
DFAST

Languages

Chinese (Native or Bilingual)
English (Full Professional)

Xuewu Cai

Director, Financial Engineering at Freddie Mac
Clarksville, Maryland, United States

Summary

• US citizen.
• A tech lead and people manager in finance business IT/financial engineering area of GSE. Managing multiple projects to help modeling loan/cohort behaviors, facilitating pricing of structured deals (credit risk transfer), estimating risk, capital and ROE, forecasting financial BS/IS and projecting DFAST simulation. Deeply engaged in Intex products and competent in structured finance deals, while implementing models with large cloud computing platforms (IBM symphony and AWS)
• A hedge fund veteran with hand-on experiences on quantitative modeling, derivative trading, risk management, high frequency and data analysis, automatic algorithm trading
• A seasoned software engineer with 20 years experiences in finance companies, proficient in Java , python, Matlab, AWS/IBM symphony, docker and many other technologies.

Experience

Freddie Mac

Director, Financial Engineering
May 2014 - Present (12 years 6 months)
McLean, Virginia, United States

• Managing projects to build modeling tools For FreddieMac CRT (Credit Risk Transfer) business line to issue STACR bonds and ACIS reinsurance policies. Focus on building the modeling tools based on FreddieMac various economic models (prepayment, default), taking advantage large scale cloud computing platform(IBM symphony, AWS), leverage Intex valuation tool to derive precise credit bond analytics and eventually produce deal transaction economic ROCC report. Managing transition from Libro to SOFR rate. CRT program in FreddieMac is protecting > 1.6 trillion mortgage loans from severe credit loss like in 2008.

* Responsible for portfolio execution/risk mamagement service which can produce risk matrices, capital benefit, post deal capital and ROE for entire FreddieMac credit risk transfer portfolios.
* Owning a key project line to forecast company PnL (income and balance sheet) and CECL. This product forecasts enterprise capital, risk appetite and provides primary support for enterprise' annual business planning. It is also the only tool for freddie mac annual DFAST, capital planning.
* Deeply involved and very knowledgeable about various structured finance vehicles (STACR, Senior/Sub, ACIS reinsurance etc.). Very familiar with IntexCalc/DealMaker and Intex cdi structure. Hands-on knowledge on Intex valuation, bond structure etc.
* First hand experience on cloud computing (IBM symphony, AWS)

Campbell and Company
Research Engineer
March 2003 - April 2014 (11 years 2 months)
Washington DC-Baltimore Area

2009- Research engineer, Portfolio and risk Team, Campbell & Company, MD
• Build volatility and correlation forecast models in Matlab using daily/intra-day data, implied volatility and tick data.
• Using various statistical methods, Monte Carlo simulation, bootstrapping methods to seek new risk factors/indicators to better predict market stress, early sign of major portfolio drawdown, potential regime switch and correlation breakdown, works are conducted in Matlab, C/C++/Java environment.
• Design and implement internal risk reporting system in Matlab (key risk numbers, factor analysis, scenario analysis, stress test, duration and margin analysis etc.) and provide timely decision support for Investment Committee.
• Provide marketing and compliance support by producing regulatory (or industry standard) documents (SEC, CFTC, OPERA etc.) in Matlab, StyleReport/Java, Panopticon.

2006-2008 Quantitative finance engineer, Campbell & Company, MD
• Built a TimeSeries Database based on FAME package using MFC/C++.
• Built cash equity DataFeed to communicate with MarketQA using MFC/C++.
• Led a team and developed a cash equity intra-day trading system using Java and Cache database (an OO database)
• Designed and developed a universal Matlab connection driver in Java to unify access to all production databases (Oracle, SQL server, Cache, QADirect etc.)

• Designed and built FX option pricing/simulation/risk engine in Visual C++/ BOOST/Quantlib.

2003-2006 Senior software engineer, Campbell & Company, MD
• Led a team to architect, design and build an electronic trading system over Java, PORTWARE and FIX engine. This multi-threading and fault-tolerant system can simultaneously execute multiple strategies and orders and trade multiple asset classes. Over 3B capital was running over this platform and saved tens of millions in slippage cost.

Aquilent

Senior Software Engineer

June 1999 - February 2003 (3 years 9 months)

OASIS (Ocean Atmosphere Sensor Integration System)
Mr. Cai ‘s role on this project includes customization of IRC for OASIS, configuration of a small Linux image to fit into PC104’s Disk-On-Chip, development of a JNI driver layer to allow Java to read/write analog and digital data from/to RTD’s DM7520 Data Acquisition board, RS-232 and Dallas’s 1-wire A/D converter.

SWAP(Sensor Web Application Prototype)
Mr Cai is a major software/hardware engineer. He deploys LinuxMZ on 6 Tri-M/IDAN PC104s, each of which is connected to either a tipping bucket rain gauge or a weather station through serial port. He also customized Linux JCOM (RXTX) through Java JNI to make specific rain gauges talk with RS232. To describe the formats of data and commands, he defined an IML (Instrument Markup Language) in XML He also incorporated dynamic sensor discovery using JXTA API.

Constellation satellites data management and visualization
. Mr. Cai conducted the initial server-side design of the system following RUP (Rational Unified Process) with the help of Rational Rose/ClearQuest. To describe the data model and hierarchy, he developed a XML parser based on JDOM and Apache/Xerces. He also selected Rhino JavaScript and built a flexible model equation engine.

Triana Web Portal
. Mr. Cai designed a AppletWrapper so that VisAGE visualization components can easily become applets. The portal was built on Apache/Tomcat platform. Mr Cai implemented visualization beans and created a taglig for HTML

authoring. He also designed an administration page that could let administrator remotely control the maximum session numbers, delete sessions, and even shutdown or restart the server. Another important achievement Mr. Cai made is a special RMI SocketFactory so that the server can still create the socket connection to the browser even the browser is behind the enterprise firewall, this solved the RMI call back through firewall problem.

Education

CAIA (Chartered Alternative Investment Analyst) Level I

University of Maryland Baltimore County
Master's Degree, Computer Science · (1997 - 1999)

Shanghai Jiao Tong University
Master of Science (M.S.), Computer Science & Engineer · (1994 - 1997)

Dalian University of Technology
Bachelor's Degree, Applied Mathematics · (1988 - 1992)